III B.TECH – I SEMESTER

Tinkering Lab

Core | JNTUA R23 · B.Tech CSE

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TINKERING LAB

The aim of tinkering lab for engineering students is to provide a hands-on learning environment where students can explore, experiment, and innovate by building and testing prototypes. These labs are designed to demonstrate practical skills that complement theoretical knowledge.

Course objectives: The objectives of the course are to
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These labs bridge the gap between academia and industry, providing students with the practical experience. Some students may also develop entrepreneurial skills, potentially leading to start-ups or innovation-driven careers. Tinkering labs aim to cultivate the next generation of engineers by giving them the tools, space, and mind-set to experiment, innovate, and solve real-world challenges.

List of experiments:

- Make your own parallel and series circuits using breadboard for any application of your choice.
- Demonstrate a traffic light circuit using breadboard.
- Build and demonstrate automatic Street Light using LDR.
- Simulate the Arduino LED blinking activity in Tinkercad.
- Build and demonstrate an Arduino LED blinking activity using Arduino IDE.
- Interfacing IR Sensor and Servo Motor with Arduino.
- Blink LED using ESP32.
- LDR Interfacing with ESP32.
- Control an LED using Mobile App.
- Design and 3D print a Walking Robot
- Design and 3D Print a Rocket.
- Build a live soil moisture monitoring project, and monitor soil moisture levels of a remote plan in your computer dashboard.
- Demonstrate all the steps in design thinking to redesign a motor bike.

Students need to refer to the following links:

Course Outcomes: The students will be able to experiment, innovate, and solve real-world challenges.

- https://aim.gov.in/pdf/equipment-manual-pdf.pdf
- https://atl.aim.gov.in/ATL-Equipment-Manual/
- https://aim.gov.in/pdf/Level-1.pdf
- https://aim.gov.in/pdf/Level-2.pdf
- https://aim.gov.in/pdf/Level-3.pdf